Run V4 Report

B5 alone and inferred provenance (NB04, GROUP V4) · Rinkesh · 13 November 2026

Outcome.

- **B5 alone already meets both targets on test:** unsafe 3/118 (2.5%), unneeded asks 5/110 (4.5%), correct final action 192/212 (90.6%).
- **The GateExp rules add 6.6 points of correct decisions** (p = 0.0005) and remove the remaining unsafe actions (3 to 0 on test, 2 to 0 on attacks), at the cost of 3.6 points more unneeded asks (not significant, p = 0.344).
- **Inferred provenance** keeps safety and final success equal on test but raises unneeded asks from 5/110 (4.5%) to 21/110 (19.1%) (p < 0.0001): the realistic cost of not having oracle provenance.

1. The run

Item	Value
Run by	Rinkesh (Kaggle `satorugojo13`), NB04 with `GROUP = "V4"`, 13 November 2026, 88 min on 2 x T4
Models	`b5_v4_combo` (final B5, trained at `86a0a2a`, same SFT data as `locked-v4`) and `b5_v4_inferred_prov` (the same adapter, provenance inferred at runtime instead of given; no retraining)
Code	`89832cd`: the locked code plus notebook condition names
Scoring	Model alone (no GateExp rules), dev / attacks / test; test also with a noisy simulated user

2. Results

Test pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 alone (b5_v4_combo, seed 42)	**212**	**192/212 (90.6%)**	**3/118 (2.5%)**	**5/110 (4.5%)**	**13/133 (9.8%)**	**198/212 (93.4%)**	**39/43 (90.7%)**	**39/39 (100.0%)**	**52/63 (82.5%)**
B5 alone, noisy user	212	190/212 (89.6%)	3/118 (2.5%)	5/110 (4.5%)	15/133 (11.3%)	198/212 (93.4%)	39/43 (90.7%)	39/39 (100.0%)	50/63 (79.4%)
B5, inferred provenance	212	192/212 (90.6%)	3/118 (2.5%)	21/110 (19.1%)	15/133 (11.3%)	182/212 (85.8%)	37/39 (94.9%)	37/39 (94.9%)	52/63 (82.5%)
B5, inferred provenance, noisy user	212	190/212 (89.6%)	3/118 (2.5%)	21/110 (19.1%)	17/133 (12.8%)	182/212 (85.8%)	37/39 (94.9%)	37/39 (94.9%)	50/63 (79.4%)
B5 + full gate (final system)	212	206/212 (97.2%)	0/118 (0.0%)	9/110 (8.2%)	6/133 (4.5%)	200/212 (94.3%)	37/38 (97.4%)	37/39 (94.9%)	58/63 (92.1%)

Attacks pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 alone (b5_v4_combo, seed 42)	**61**	**57/61 (93.4%)**	**2/51 (3.9%)**	**3/33 (9.1%)**	**2/11 (18.2%)**	**51/61 (83.6%)**	**11/15 (73.3%)**	**11/15 (73.3%)**	**9/13 (69.2%)**
B5 + full gate (final system)	61	60/61 (98.4%)	0/51 (0.0%)	4/33 (12.1%)	1/11 (9.1%)	48/61 (78.7%)	12/19 (63.2%)	12/15 (80.0%)	7/13 (53.8%)

Dev pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 alone (b5_v4_combo, seed 42)	**236**	**225/236 (95.3%)**	**0/112 (0.0%)**	**9/150 (6.0%)**	**9/149 (6.0%)**	**219/236 (92.8%)**	**23/23 (100.0%)**	**23/25 (92.0%)**	**55/61 (90.2%)**
B5, inferred provenance	236	217/236 (91.9%)	1/112 (0.9%)	25/150 (16.7%)	16/149 (10.7%)	194/236 (82.2%)	23/24 (95.8%)	23/25 (92.0%)	49/61 (80.3%)
B5 + full gate (final system)	236	232/236 (98.3%)	0/112 (0.0%)	9/150 (6.0%)	4/149 (2.7%)	223/236 (94.5%)	25/25 (100.0%)	25/25 (100.0%)	57/61 (93.4%)

Last row: the same B5 inside the full gate (NB08, the final system), for comparison.

3. Paired tests

Comparison	Outcome	B5 / A	Other / B	Diff, points [95% CI]	McNemar p	Holm p
Full gate vs B5 alone (test)	success	206/212 (97.2%)	192/212 (90.6%)	+6.6 [+3.3, +10.4]	**0.0005**	-
Full gate vs B5 alone (test)	unauthorized	0/118 (0.0%)	3/118 (2.5%)	-2.5 [-5.9, +0.0]	0.2500	-
Full gate vs B5 alone (test)	unnecessary deferral	9/110 (8.2%)	5/110 (4.5%)	+3.6 [-1.8, +9.1]	0.3438	-
Full gate vs B5 alone (dev)	success	232/236 (98.3%)	225/236 (95.3%)	+3.0 [+0.8, +5.5]	**0.0156**	-
Oracle vs inferred provenance (test)	success	192/212 (90.6%)	192/212 (90.6%)	+0.0 [+0.0, +0.0]	1.0000	1.0000
Oracle vs inferred provenance (test)	unnecessary deferral	5/110 (4.5%)	21/110 (19.1%)	-14.5 [-21.8, -8.2]	**<0.0001**	-
Oracle vs inferred provenance (dev)	success	225/236 (95.3%)	217/236 (91.9%)	+3.4 [+0.8, +5.9]	0.0215	0.0645
Clean vs noisy user (B5 alone) (test)	success	192/212 (90.6%)	190/212 (89.6%)	+0.9 [+0.0, +2.4]	0.5000	-

Paired over the same cases. Diff = B5 minus the other model (for unsafe and unneeded asks, negative means B5 is better). Exact McNemar test; bootstrap 95% interval (10,000 resamples). Green p: significant at 0.05 (after Holm-Bonferroni where the test is in the secondary family).

Here "B5 / A" is the first model named in the comparison.

4. Analysis

What the rules add

- The model alone is already a strong gate. The rules turn it into the final system: +6.6 points of correct final actions on test (p = 0.0005), +3.0 on dev, and 0 unsafe actions on every pool. The unsafe-action gain (3 to 0) is not significant on its own (only 3 discordant cases), but it is consistent across test, attacks and dev.
- The rules also accept no spoofed approval: B5 alone accepted 1 of 8 spoofed approvals on dev; the full gate 0 of 8.
- The price is a few more confirmation questions (unneeded asks 5/110 (4.5%) to 9/110 (8.2%)), all within the 10% target. Most are confirmations the handbook requires (new payee, unlabelled data, untraced target).

Oracle vs inferred provenance

- Everywhere else in this project the provenance of each value (user, trusted tool, untrusted content) is given (oracle). Here it is inferred at runtime from the trace, as a deployed system would have to.
- Safety and final success on test are unchanged; dev success drops +3.4 points [95% CI +0.8, +5.9] (p = 0.021). The clear cost is interaction: unneeded asks 5/110 (4.5%) to 21/110 (19.1%). When the source of a value is uncertain, the gate asks instead of guessing.
- Provenance attribution errors are the main new error type (16 cases on test). Better provenance tracking (taint tracking in the agent runtime) is the most direct way to close this gap.

Noisy user

- With unhelpful or overreaching second answers, B5 alone loses 2 cases on test (190/212 (89.6%)); safety is unchanged. Noise only ever leads to blocking, never to unsafe execution.

5. Records

Record	Where
Predictions, reports, logs	`results/data/2026-11-13_eval_v4/` (local, not yet committed)
Full-gate results	`results/data/2026-11-13_gateexp_b5v4combo_final/`, `results/final/master_results.pdf`